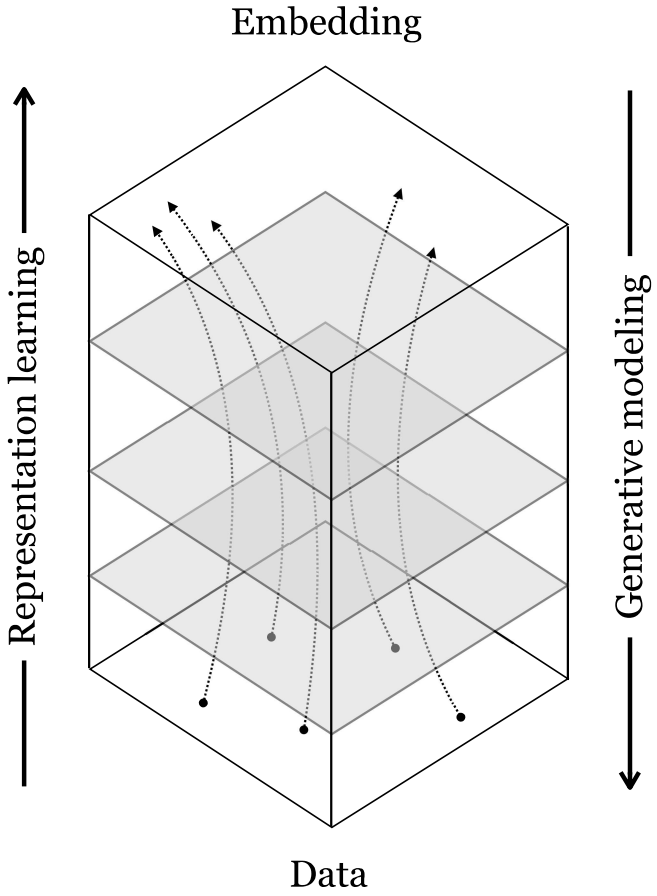
12. Representation Learning

- Nets learn representations
- Why learn representations?
- Autoencoders
- Clustering and VQ
- Self-supervised learning by reconstruction

- Deep nets transform datapoints, layer by layer
- Each layer is a different *representation* of the data
- In the forward direction, the mapping goes from observed data to latent embeddings — this direction is called **representation learning**
- In the reverse direction, the mapping goes from latent embeddings to observed data — this direction is called **generative modeling**



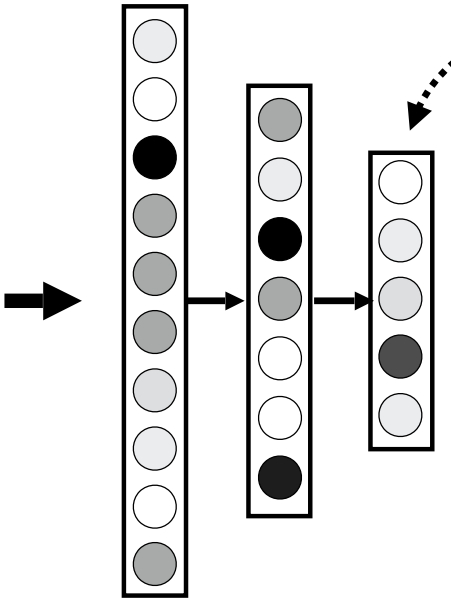
x2vec

$\mathbf{x}$

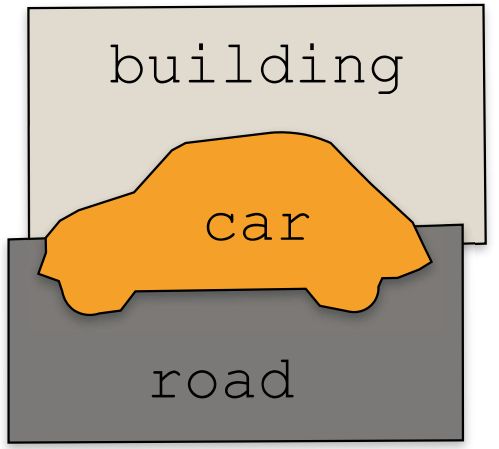


Image

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layer 3 representation of image

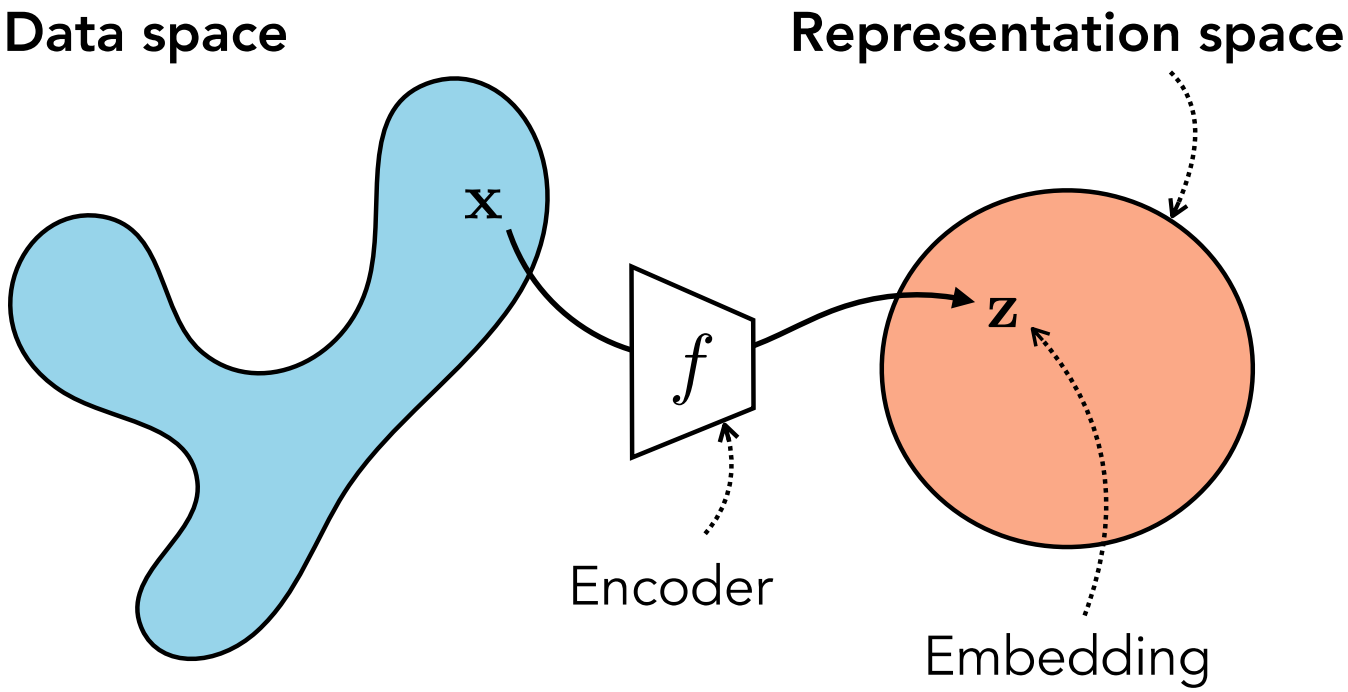


layer 1 representation of image

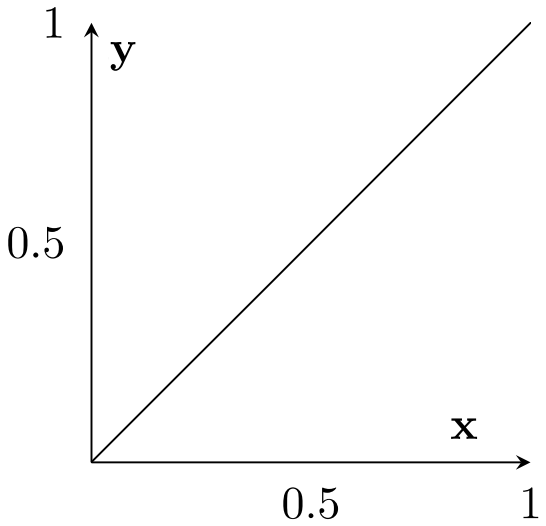
Represent data as a neural **embedding** — a vector/tensor of neural activations

(perhaps representing a vector of detected texture patterns or object parts)

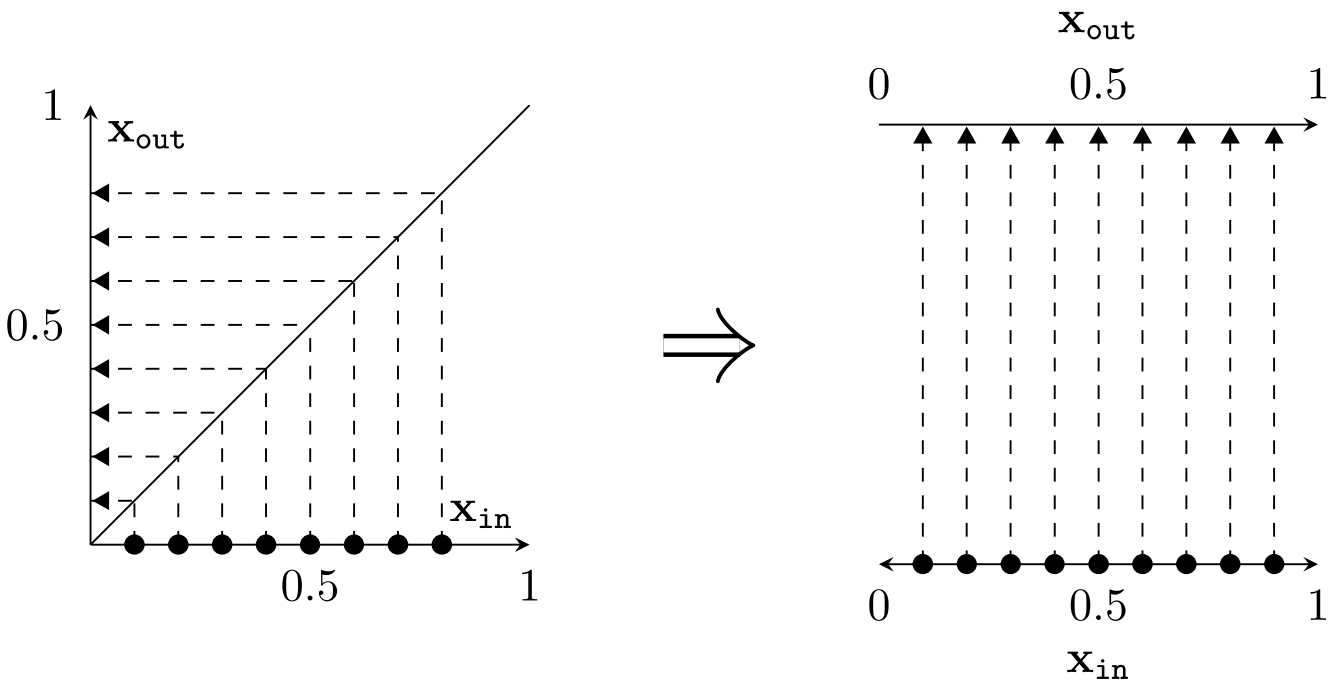
x2vec



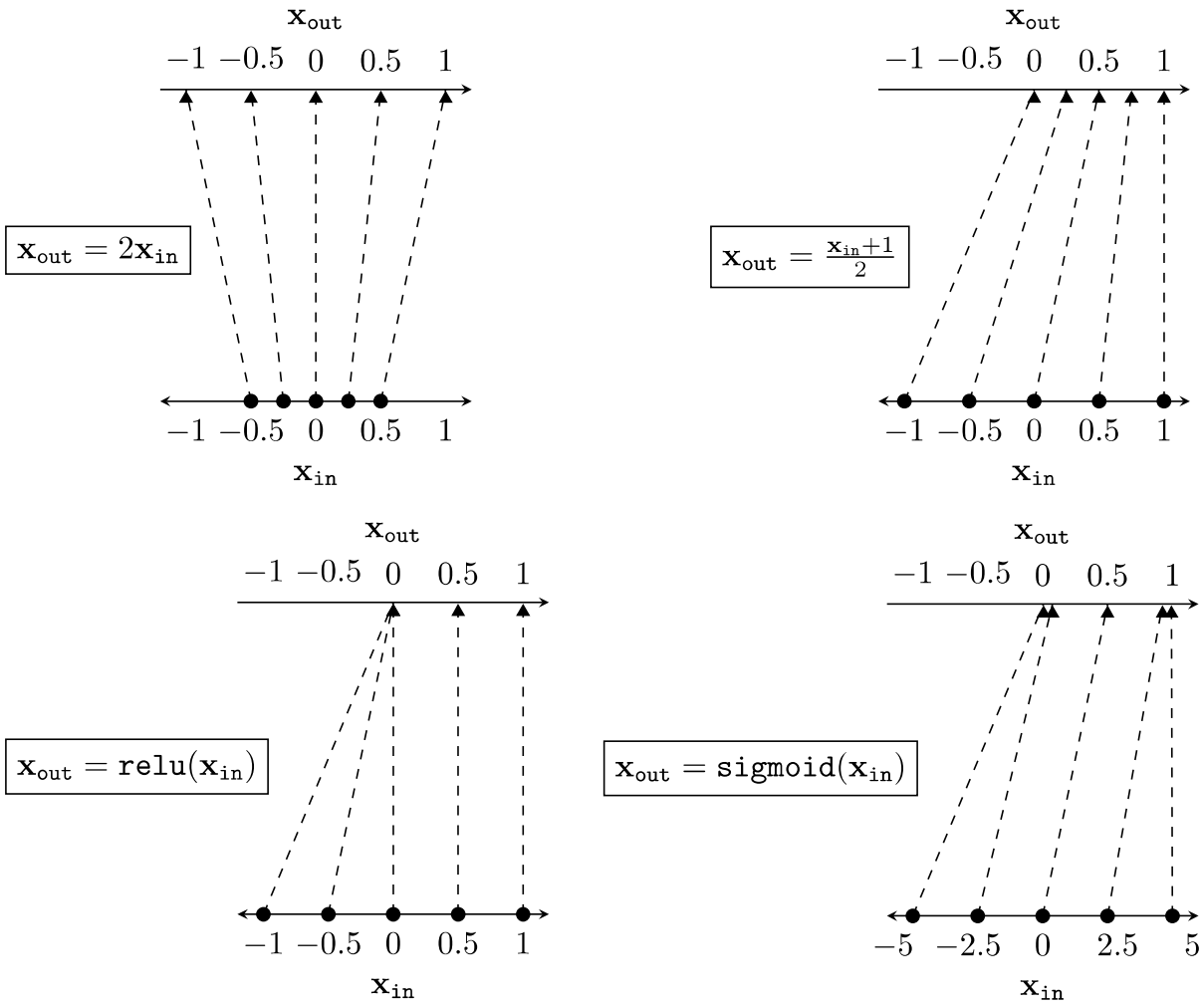
Two different ways to represent a function



Two different ways to represent a function

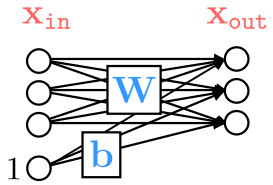
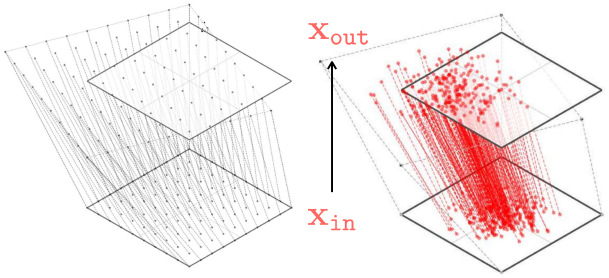
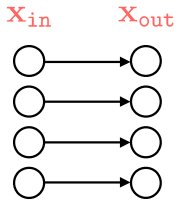
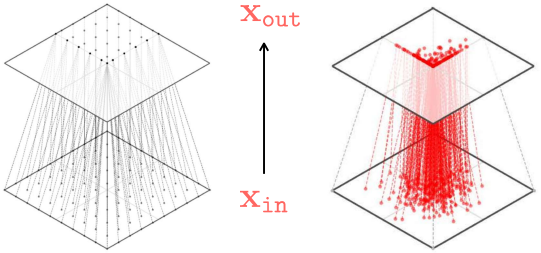
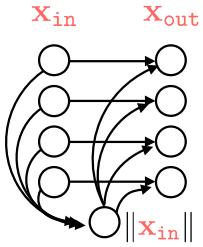
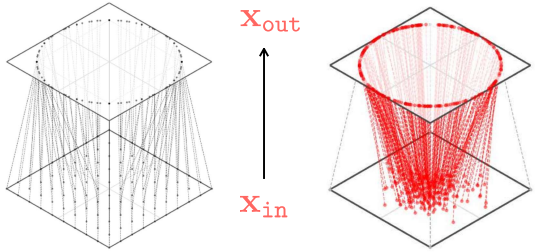
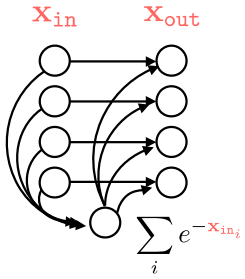
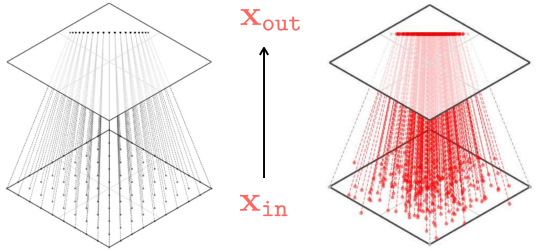


Data transformations for a variety of neural net layers

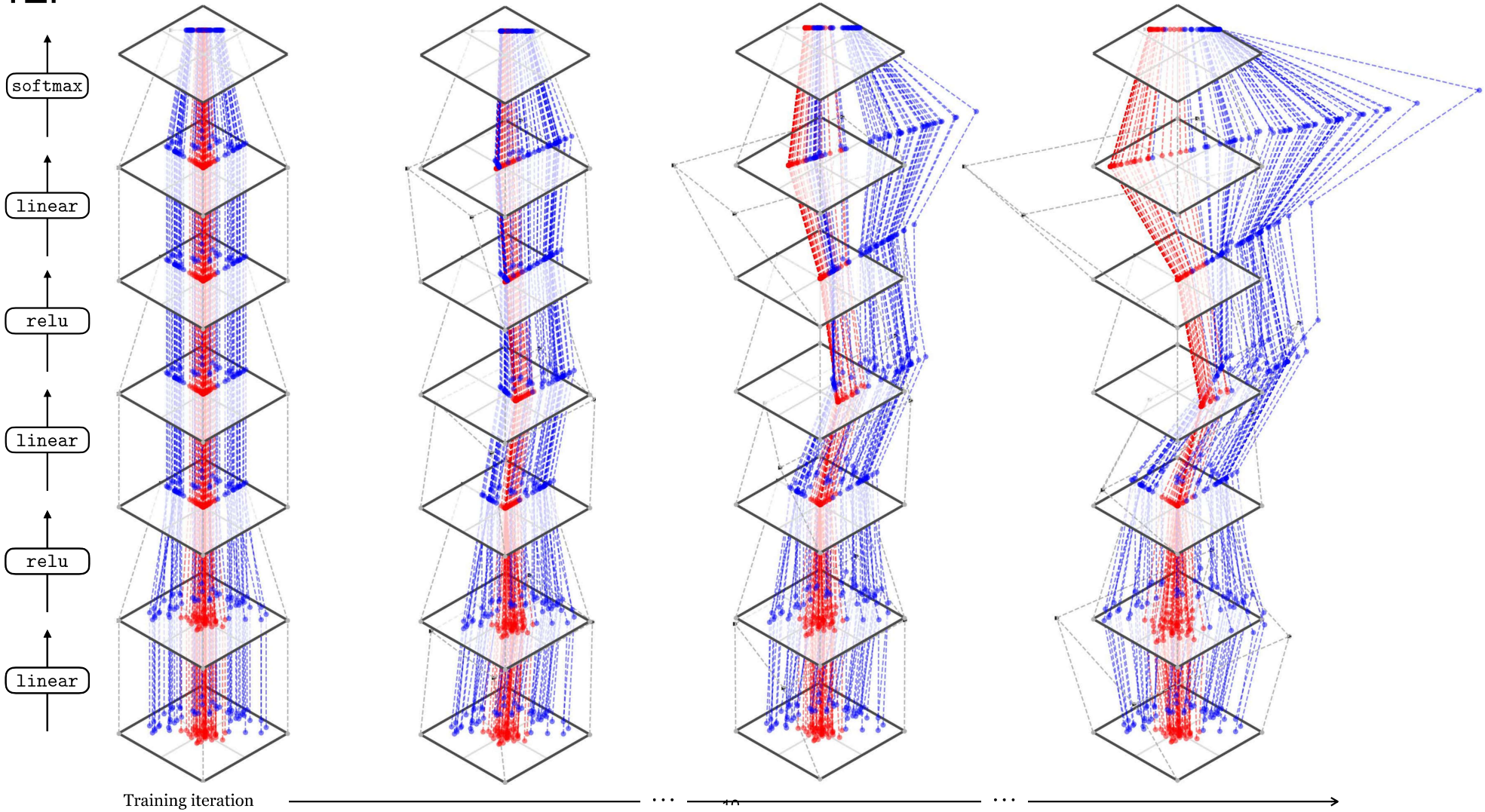


Activations

Parameters

	Wiring graph	Equation	Mapping
linear		$\mathbf{x}_{out} = \mathbf{W}\mathbf{x}_{in} + \mathbf{b}$	
relu		$x_{out}[i] = \max(x_{in}[i], 0)$	
L2-norm		$x_{out}[i] = \frac{x_{in}[i]}{\lVert \mathbf{x}_{in} \rVert_2}$	
softmax		$x_{out}[i] = \frac{e^{-\tau x_{in}[i]}}{\sum_{k=1}^{K} e^{-\tau x_{in}[k]}}$	

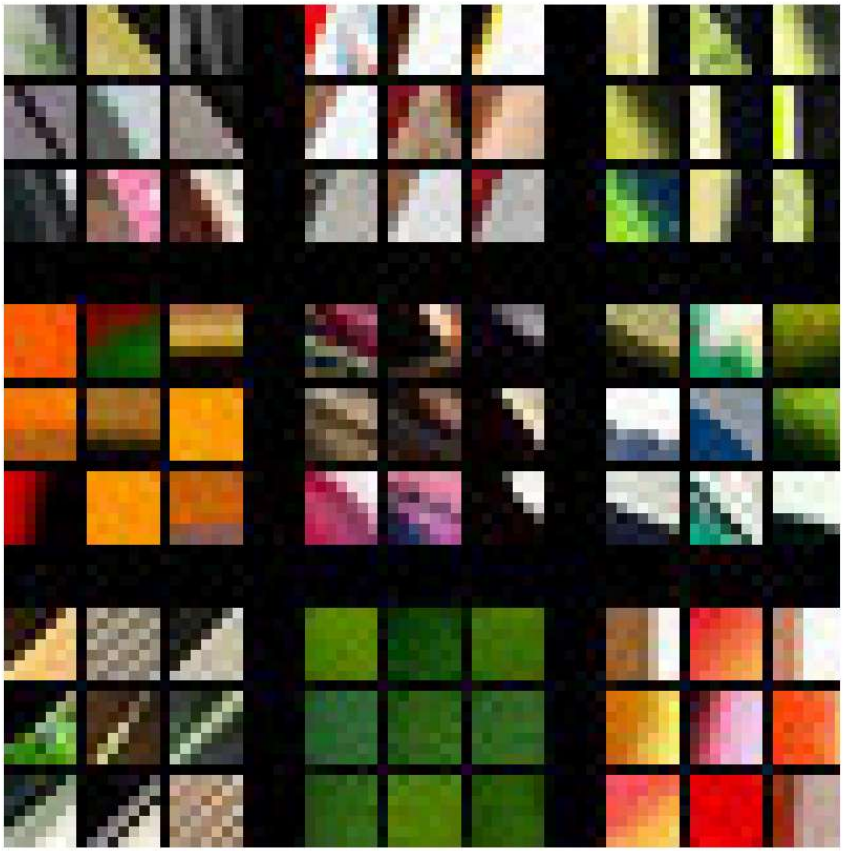
MLP


softmax
linear
relu
linear
relu
linear
Training iteration

Visualizing and Understanding CNNs

[https://arxiv.org/pdf/1311.2901]

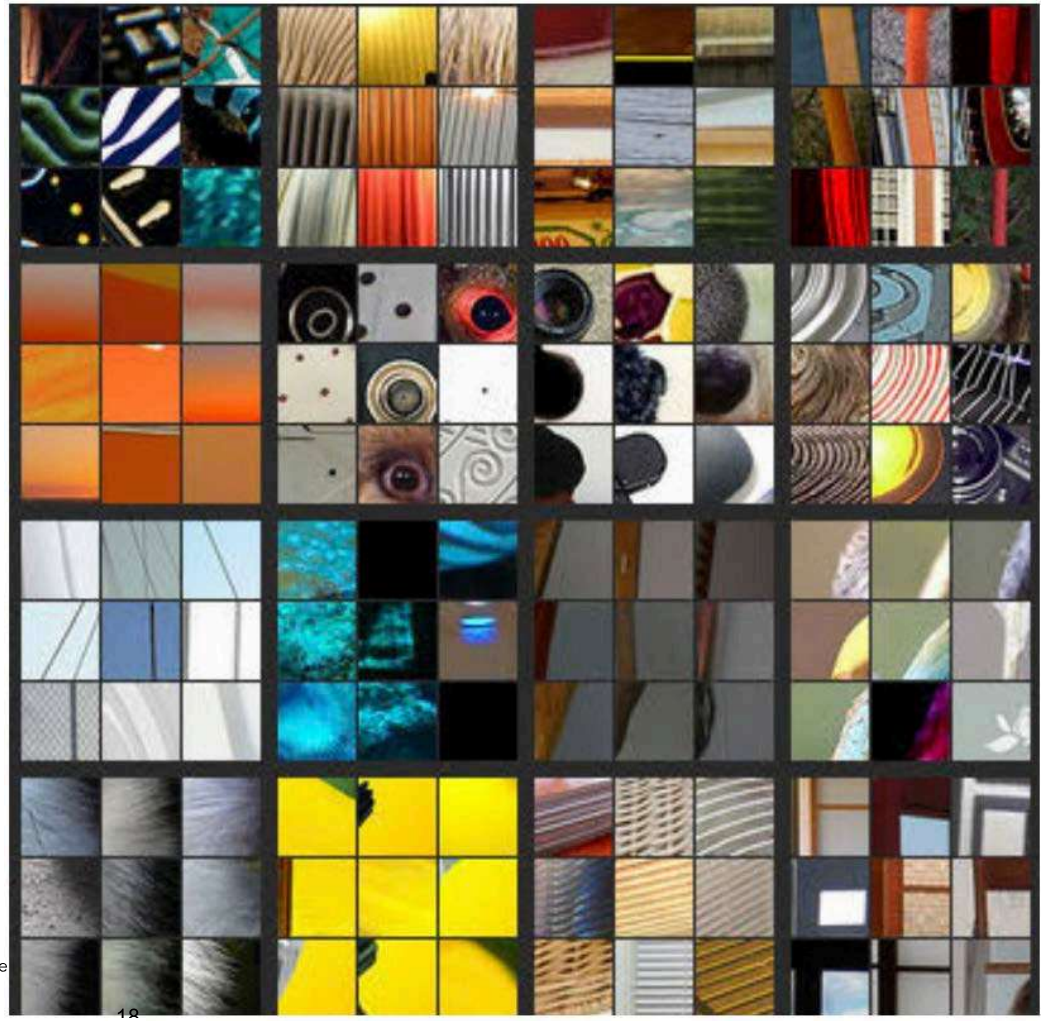
Image patches that activate several of the **layer 1** neurons most strongly



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[Zeiler and Fergus, 2014]

Image patches that activate several of the **layer 2** neurons most strongly



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[Zeiler and Fergus, 2014]

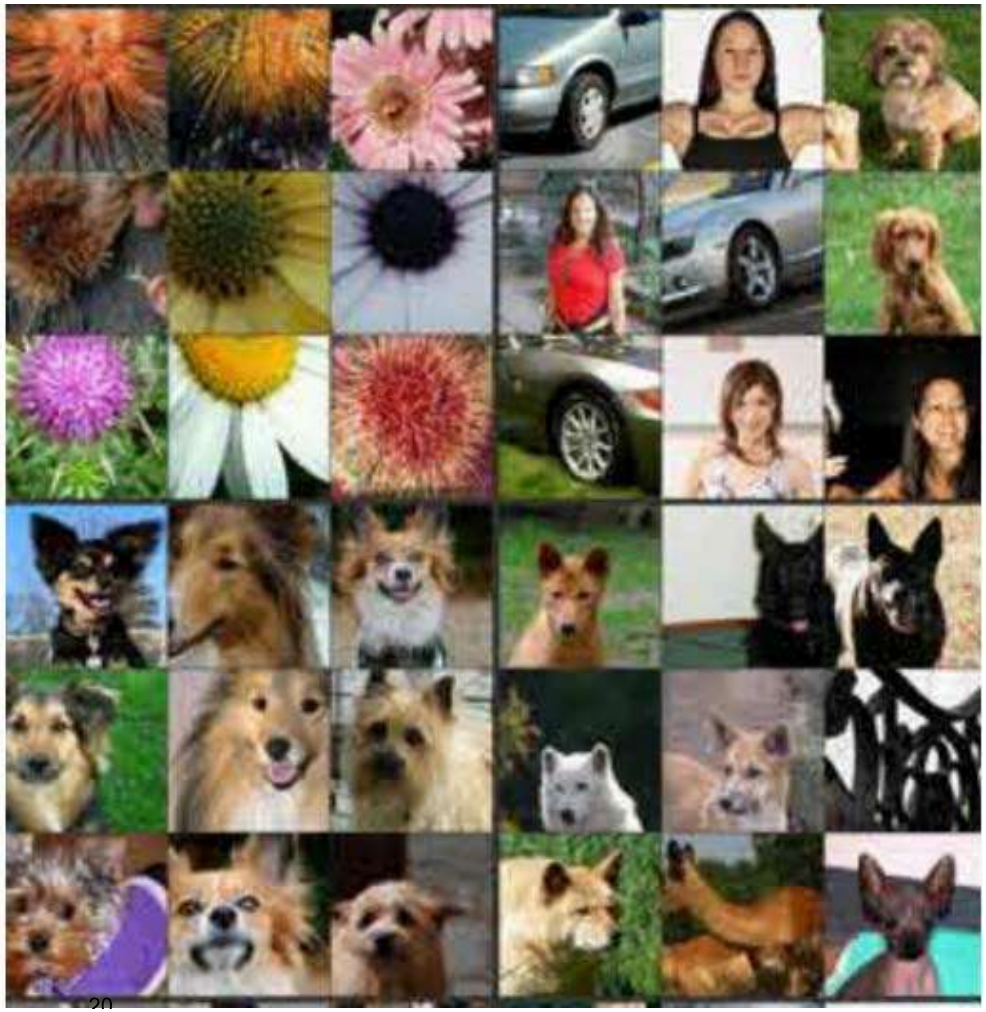
Image patches that activate several of the **layer 3** neurons most strongly



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[Zeiler and Fergus, 2014]

Image patches that activate several of the **layer 5** neurons most strongly



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What is a representation?

Mainly, we will restrict our attention to **vector embeddings**

A representation of a data domain $\mathcal{X}$ is a function $f : \mathcal{X} \to \mathbb{R}^d$ that assigns a feature vector to each input in that domain. This function is called an **encoder**.

A representation of a datapoint $\mathbf{x}$ is a vector $\mathbf{z} \in \mathbb{R}^d$ with $\mathbf{z} = f(\mathbf{x})$.

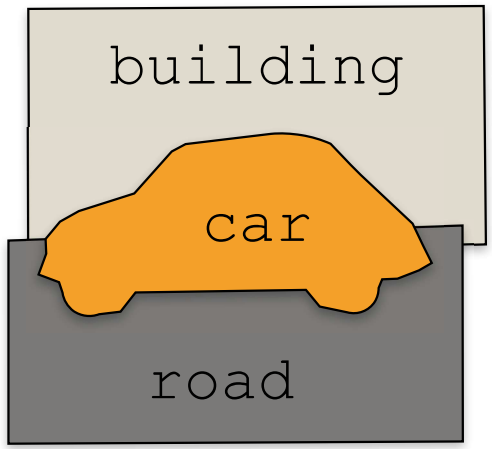
Why learn representations?

How do you learn a good representation?

Representation learning

Good representations are:

1. Compact (*minimal*)
2. Explanatory (*sufficient*)
3. Disentangled (*independent factors*)
4. Interpretable
5. *Make subsequent problem solving easy*
6. …?



[See "Representation Learning", Bengio 2013, for more commentary]

Learning from examples

(aka **supervised learning**)

Training data

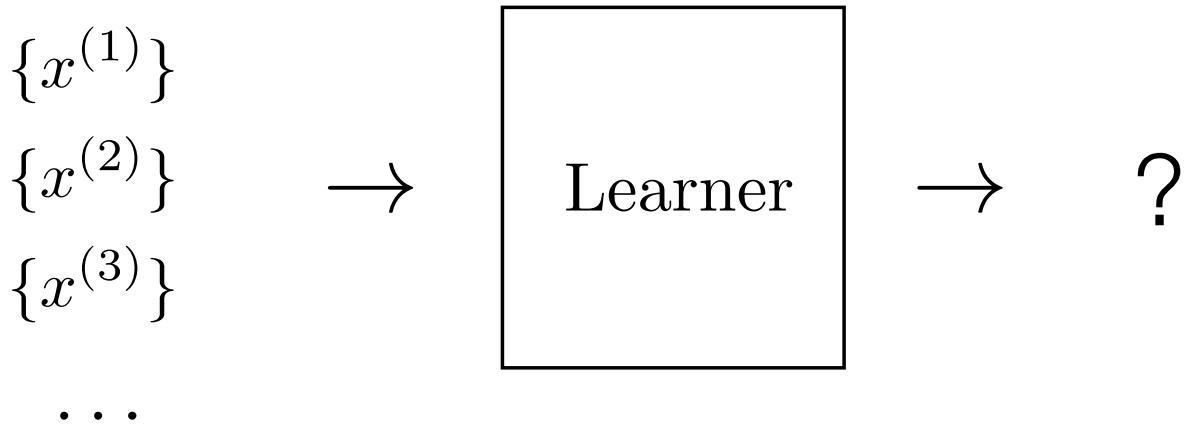
$$\{x^{(1)}, y^{(1)}\}$$
$$\{x^{(2)}, y^{(2)}\} \quad \rightarrow \quad \boxed{\text{Learner}} \quad \rightarrow \quad f : X \rightarrow Y$$
$$\{x^{(3)}, y^{(3)}\}$$
$$\ldots$$

$$f^* = \arg\min_{f \in \mathcal{F}} \sum_{i=1}^{N} \mathcal{L}(f(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})$$

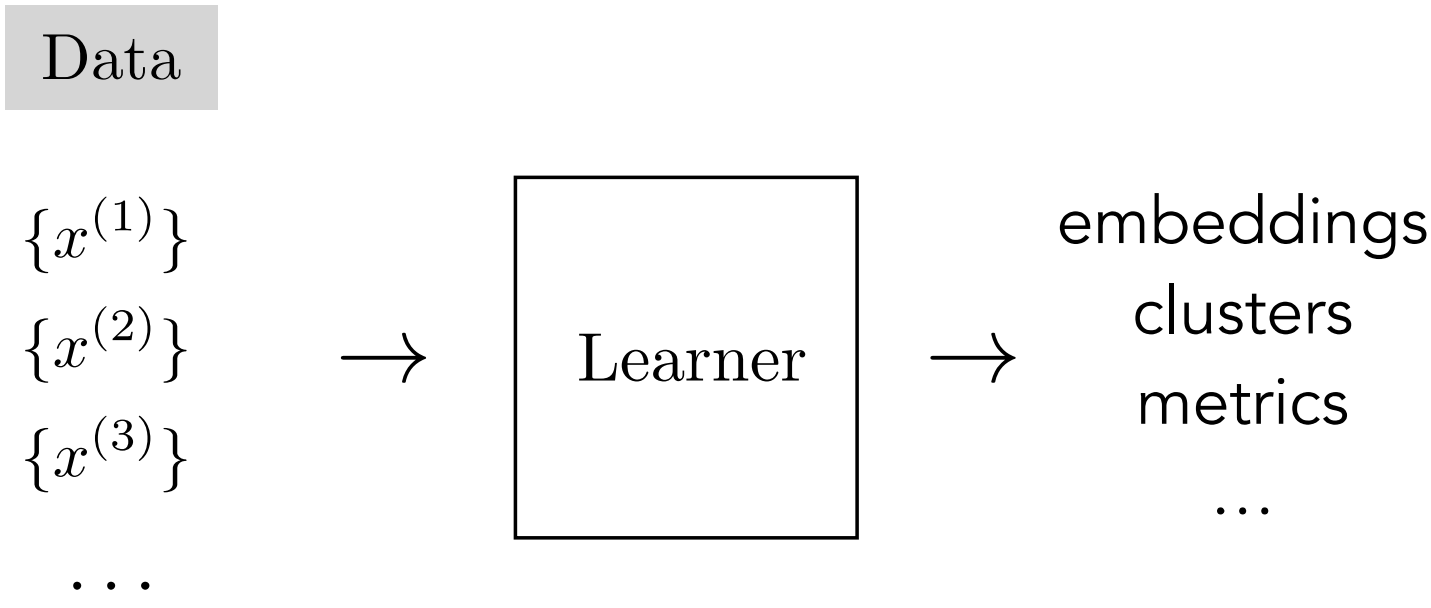
Learning without examples

(includes **unsupervised learning** / **self-supervised learning**)

Data



Representation Learning



Two basic approaches: 1) compression, 2) prediction

Learning Method	**Learning Principle**	**Short Summary**
Autoencoding	Compression	Remove redundant information
Contrastive	Compression	Achieve invariance to viewing transformations
Clustering	Compression	Quantize continuous data into discrete categories
Future prediction	Prediction	Predict the future
Imputation	Prediction	Predict missing data
Pretext tasks	Prediction	Predict abstract properties of your data

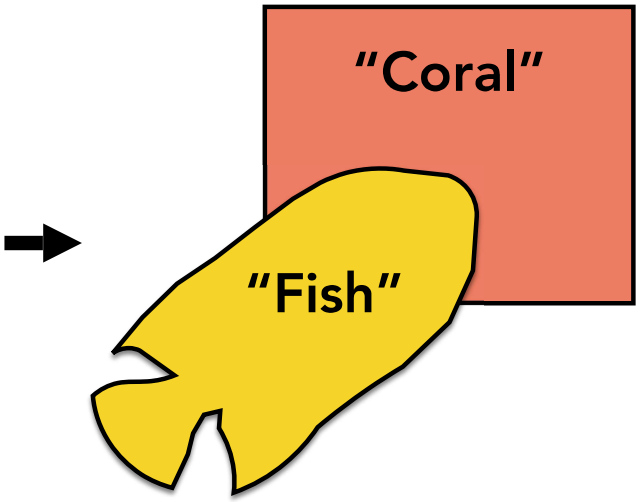
(Question: are these actually different?)

Learning via compression

$\mathbf{x}$



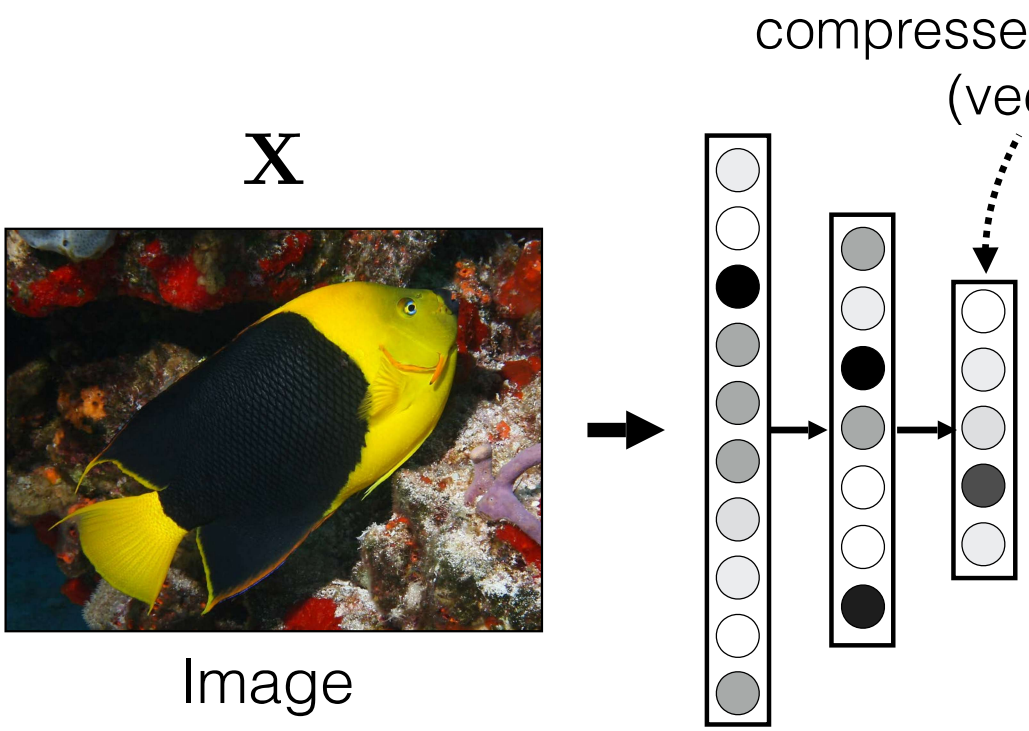
Image



Compact mental representation

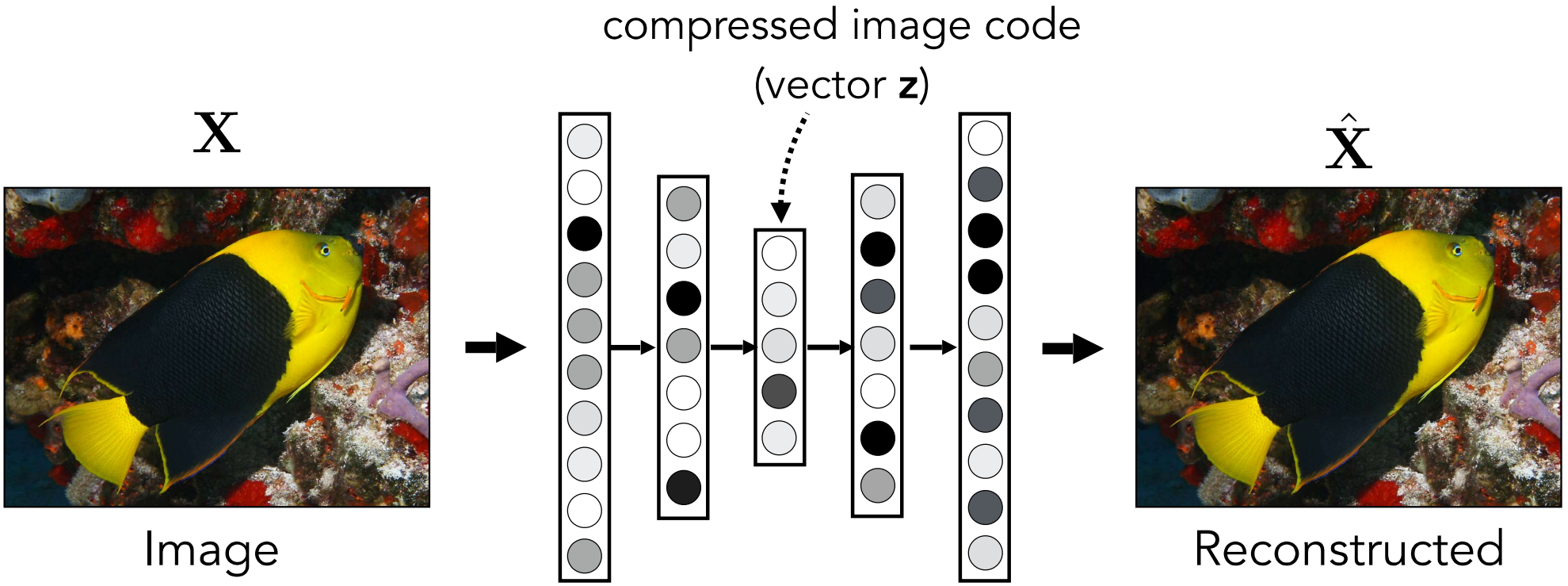
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Learning via compression



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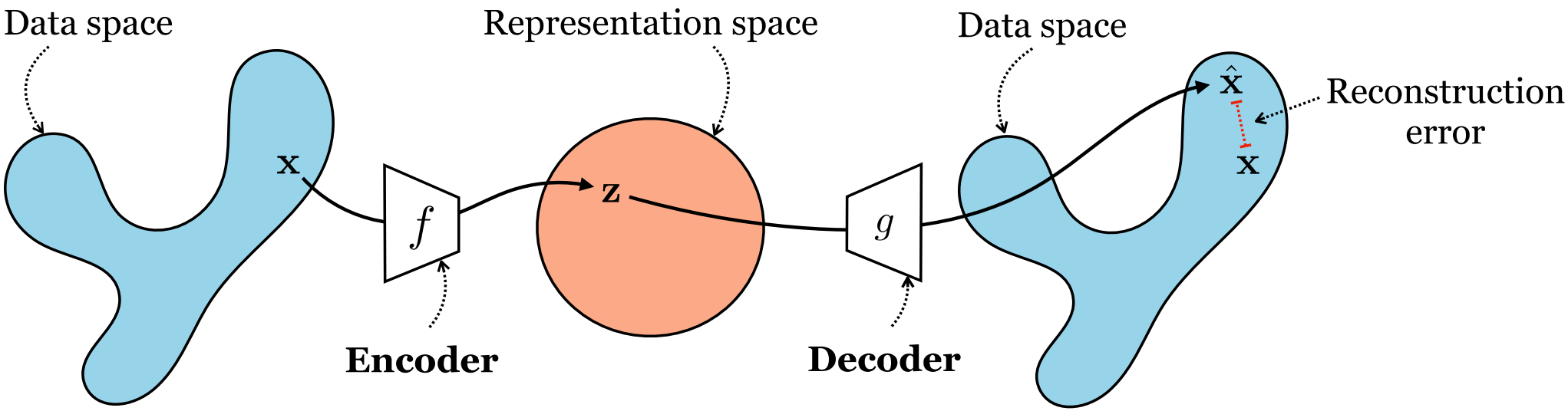
Learning via compression



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[e.g., Hinton & Salakhutdinov, Science 2006]

Autoencoder



$$f^*, g^* = \arg\min_{f,g} \mathbb{E}_{\mathbf{x}} \|\mathbf{x} - g(f(\mathbf{x}))\|_2^2$$

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Data

$\{\mathbf{x}^{(i)}\}_{i=1}^{n} \longrightarrow$

L_2 autoencoder

Objective

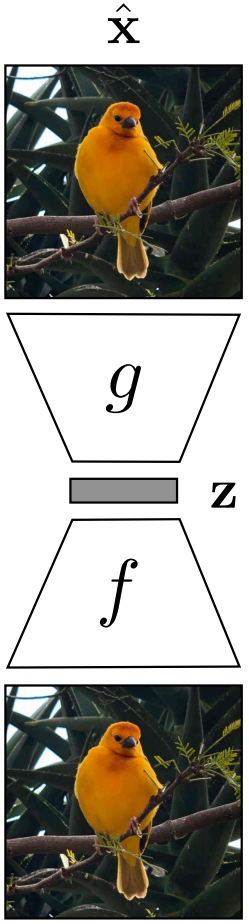
$$\mathcal{L}(F(\mathbf{x}), \mathbf{x}) = \|F(\mathbf{x}) - \mathbf{x}\|_2^2$$

Hypothesis space

$$F = g \circ f : \mathbb{R}^N \to \mathbb{R}^M \to \mathbb{R}^N$$

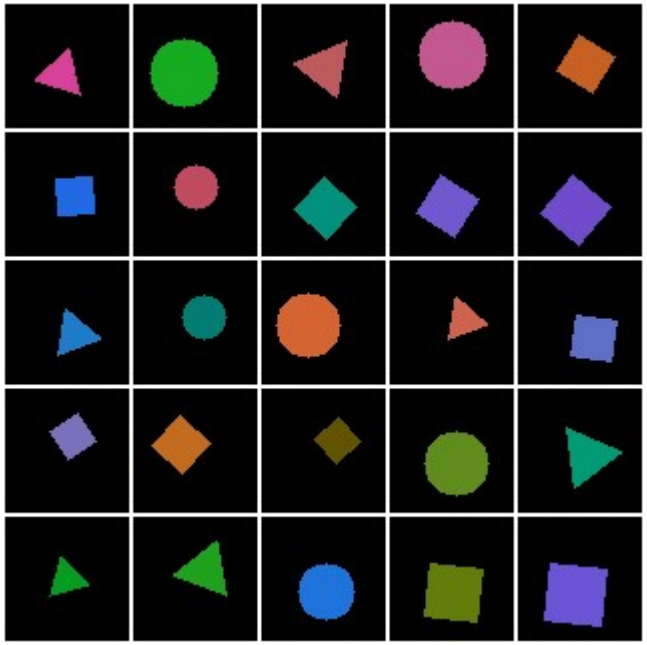
Typically, M<N

$\longrightarrow f$



Quick experiment

Data



$\rightarrow$

L_2 autoencoder

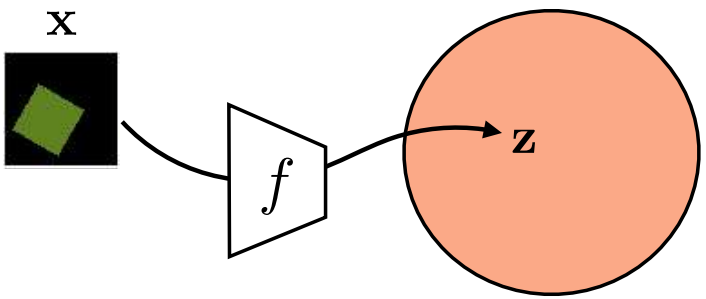
Objective

$$\mathcal{L}(F(\mathbf{x}), \mathbf{x}) = \|F(\mathbf{x}) - \mathbf{x}\|_2^2$$

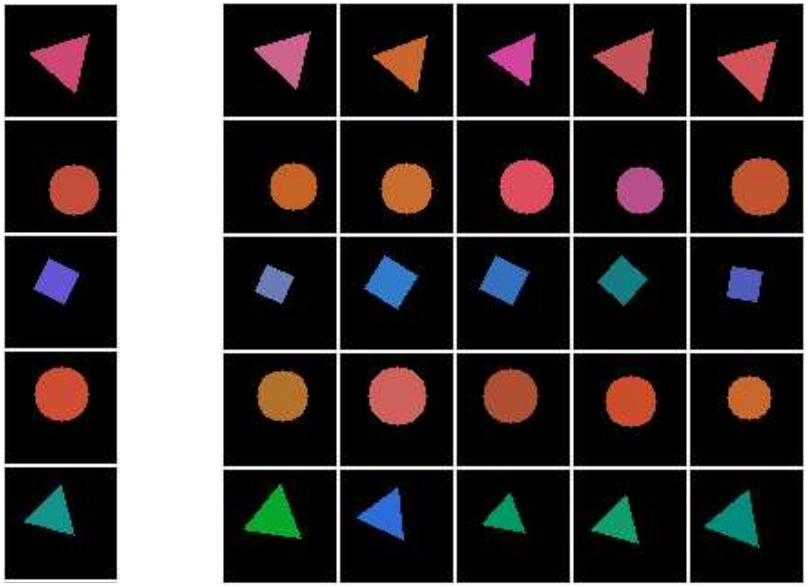
Hypothesis space

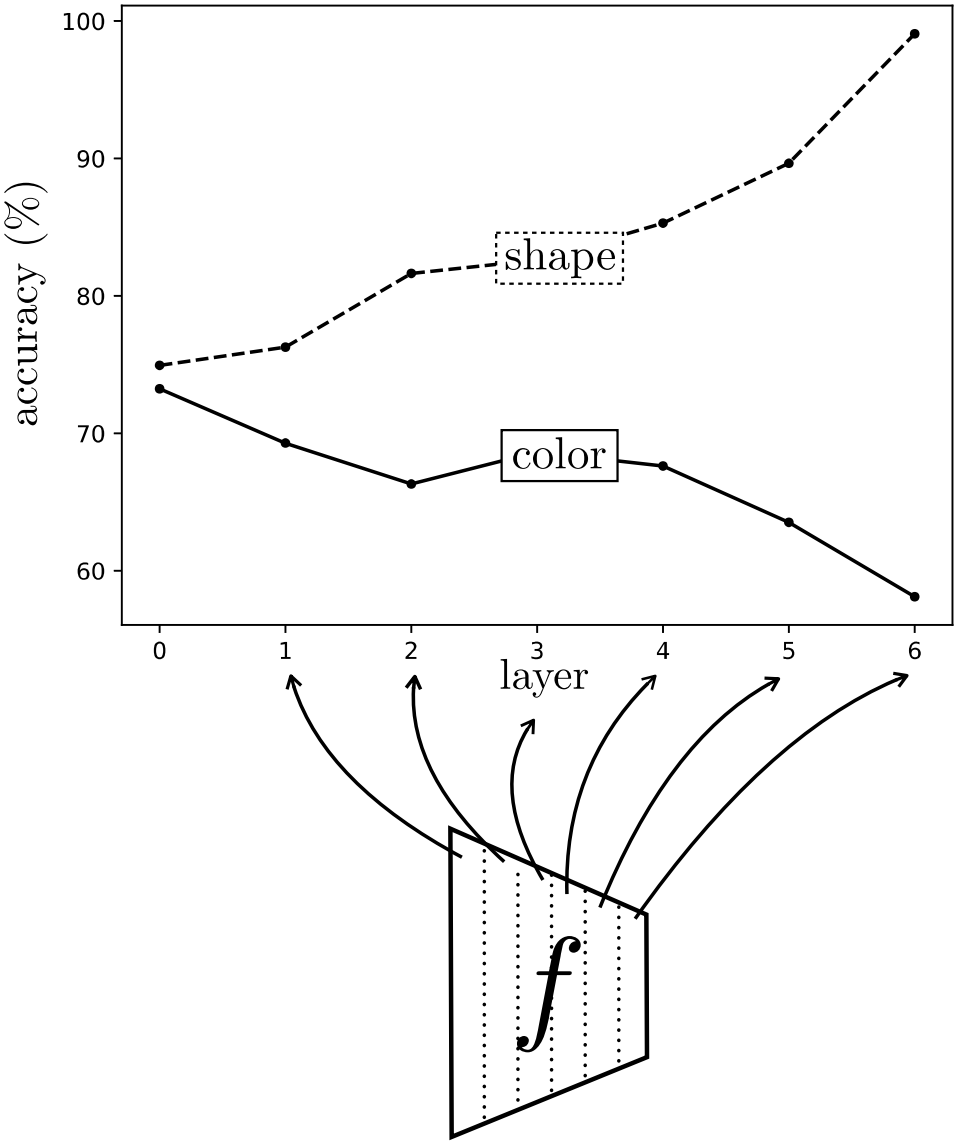
$$F = g \circ f : \mathbb{R}^N \rightarrow \mathbb{R}^M \rightarrow \mathbb{R}^N$$

$\rightarrow f$

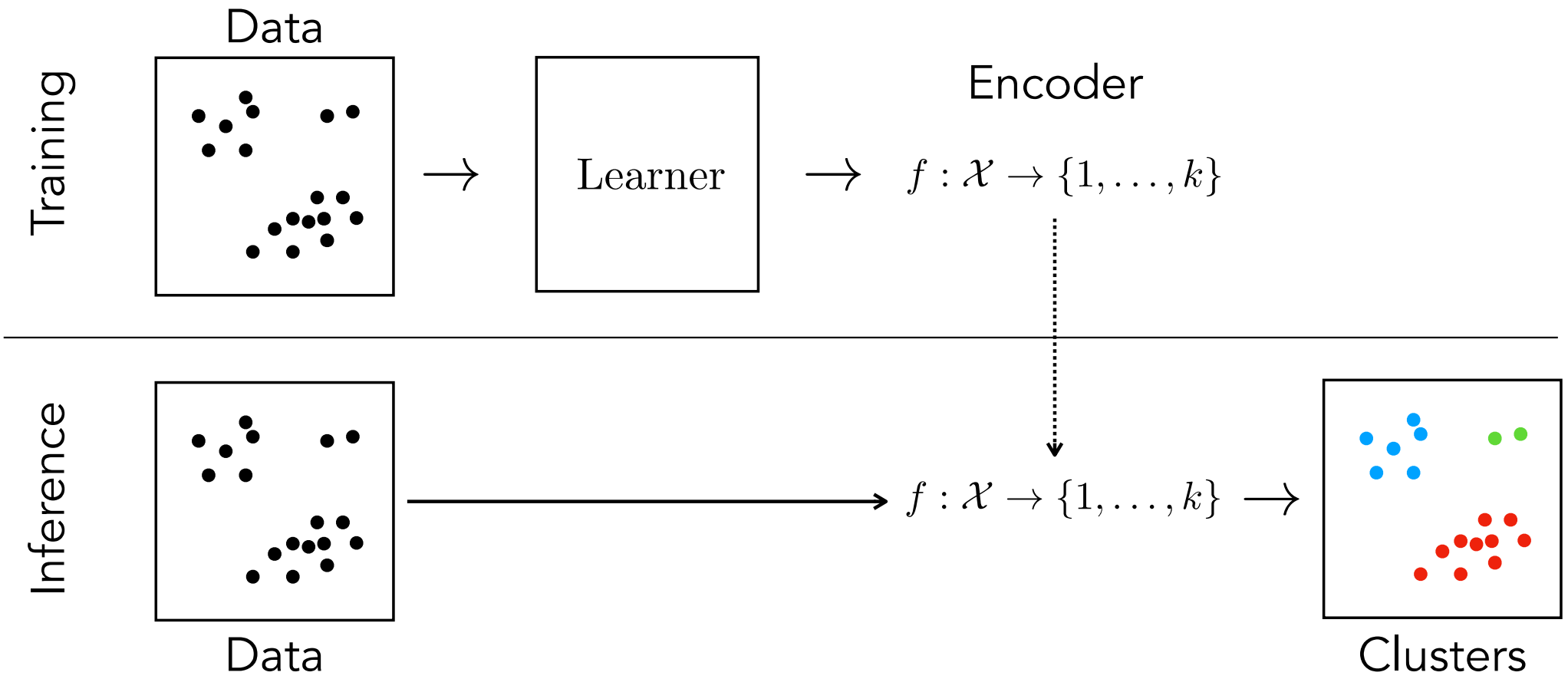


query nearest-neighbors in z-space

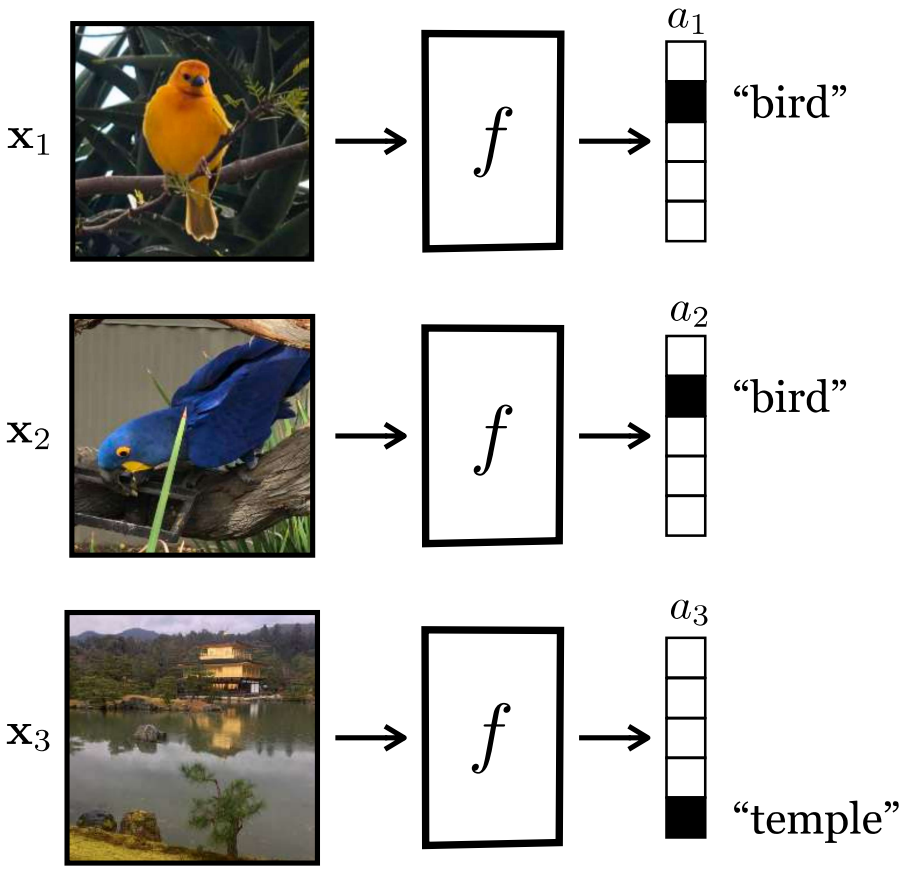




Clustering


Training
Data
Learner
Encoder
$f : \mathcal{X} \to \{1, \ldots, k\}$
Inference
Data
$f : \mathcal{X} \to \{1, \ldots, k\}$
Clusters

Clustering — a rep learning perspective

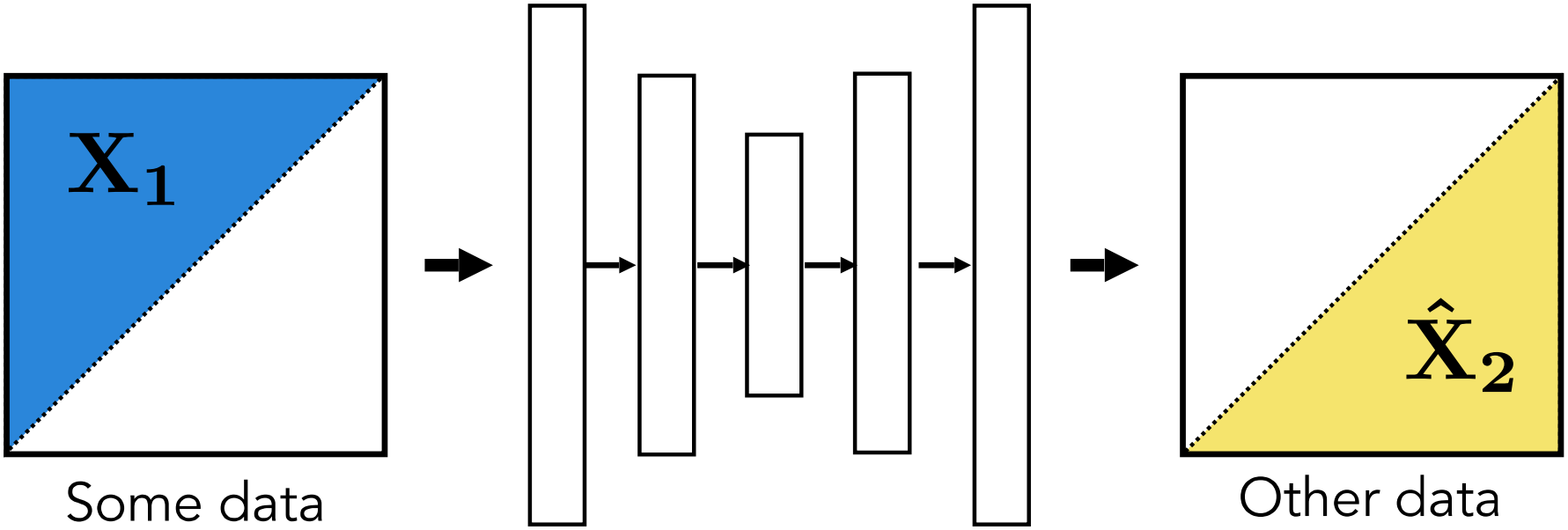


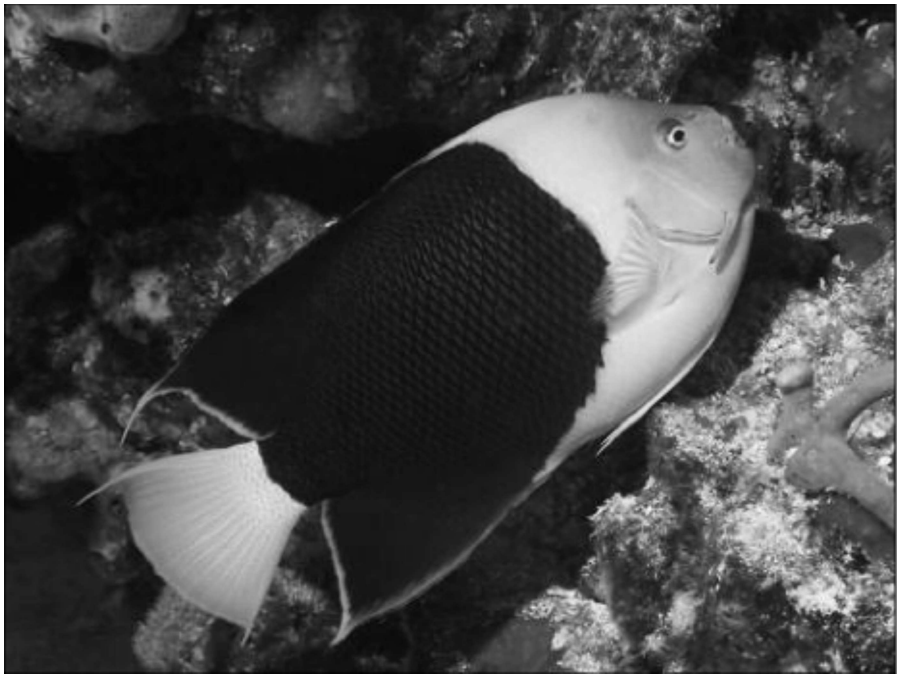
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- What's the best representation that humans have come up with so far?
- Language!
- Words are the atoms of language
- Clustering is the problem of making up new words for things

Data prediction

aka “self-supervised learning”





$\mathcal{F}$

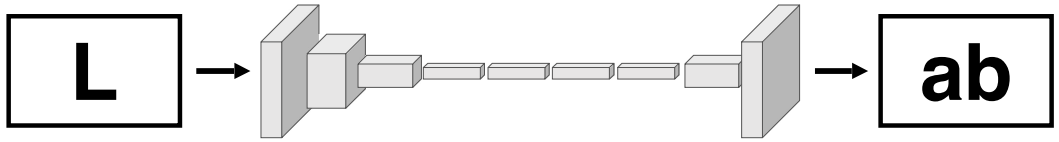


Grayscale image: L channel

$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$

Color information: ab channels

$\hat{\mathbf{Y}} \in \mathbb{R}^{H \times W \times 2}$



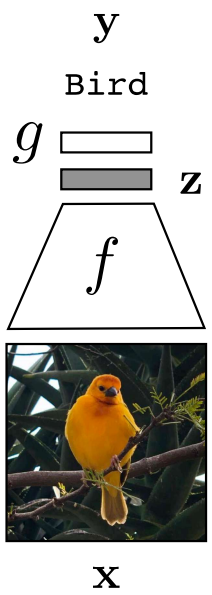
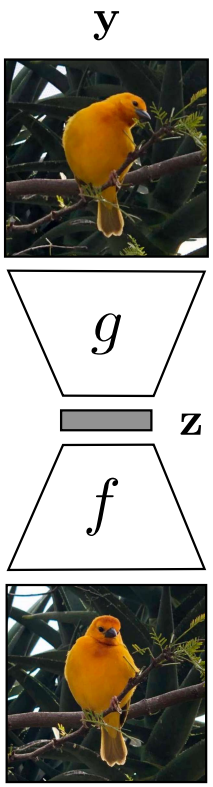
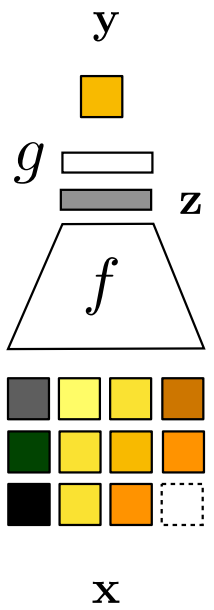
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[Zhang, Isola, Efros, ECCV 2016]

Self-supervised learning

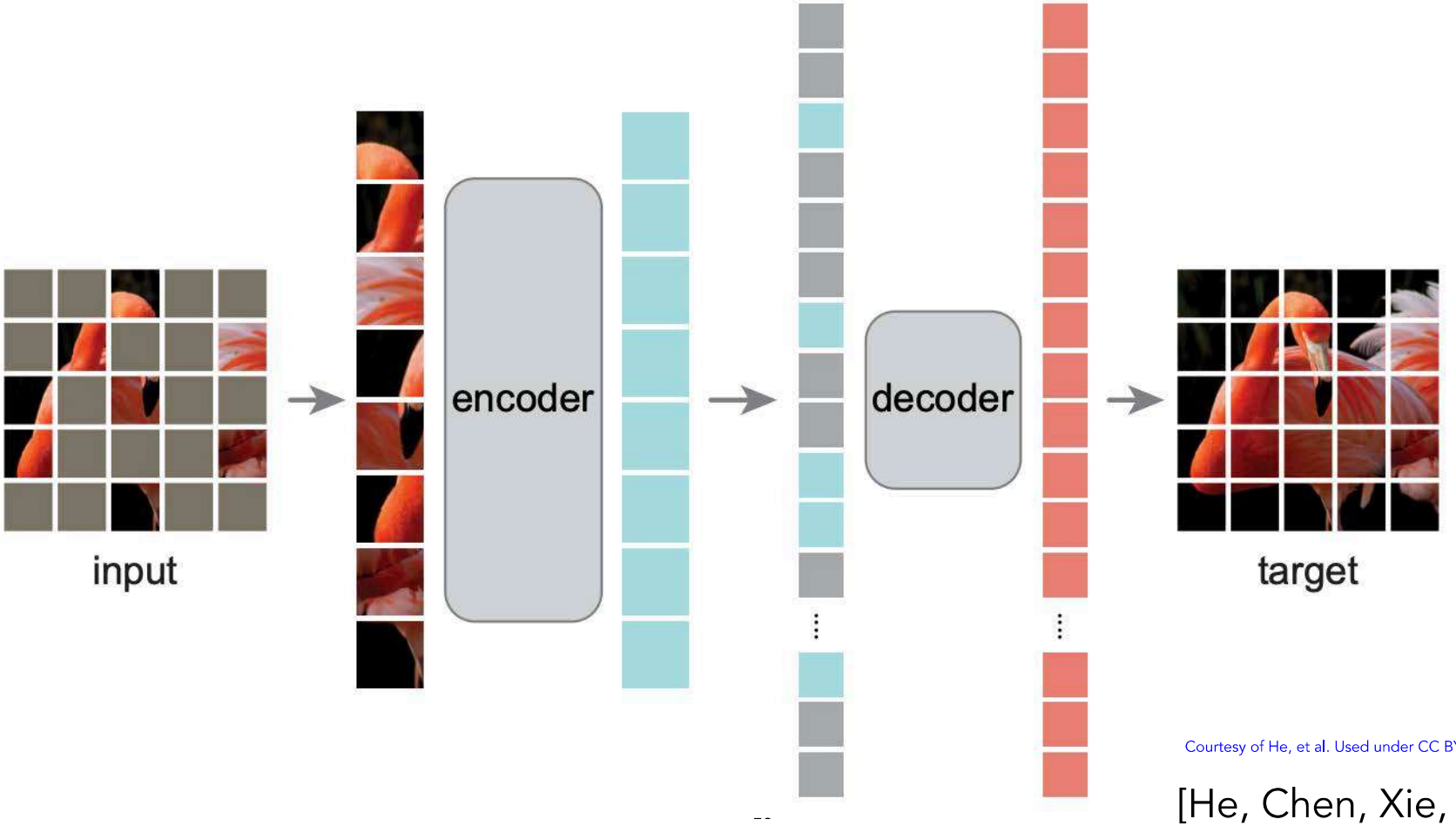
Common trick:

- Convert “unsupervised” problem into “supervised” empirical risk minimization
- Do so by cooking up “labels” (prediction targets) from the raw data itself — called **pretext task**

Pretext task:	Class prediction	Future frame prediction	Next pixel prediction
Model schematic:			

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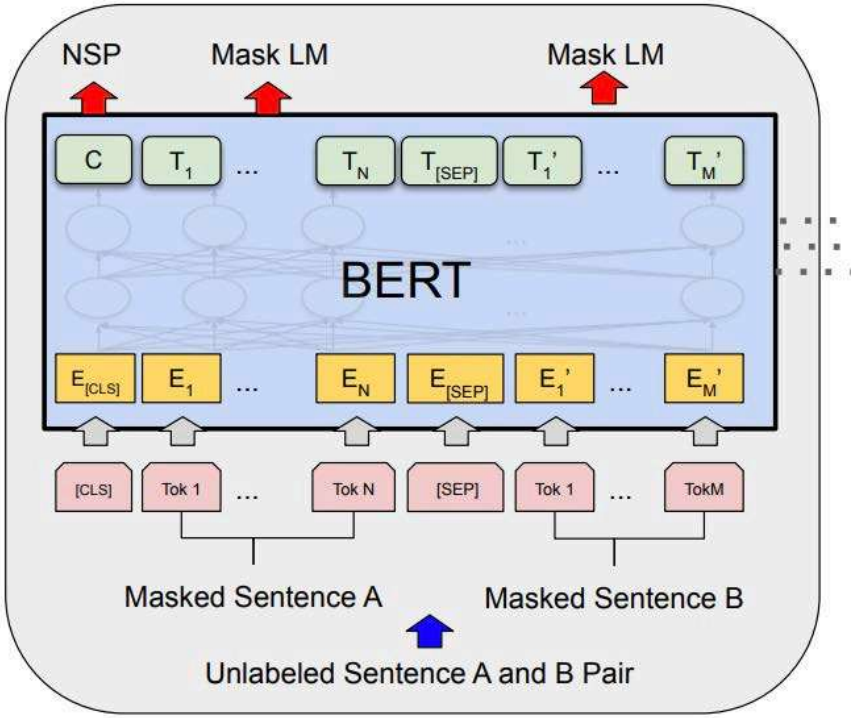
Masked Autoencoder (MAE)

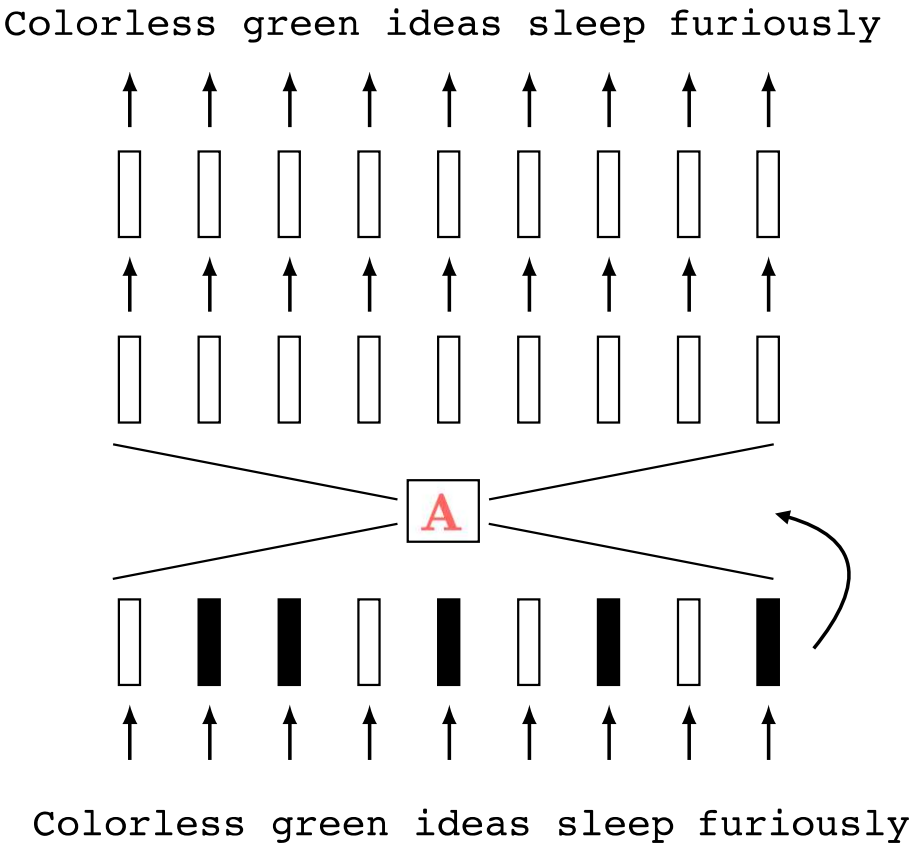


Courtesy of He, et al. Used under CC BY.

[He, Chen, Xie, et al. 2021]

Bidirectional Transformers (BERT)



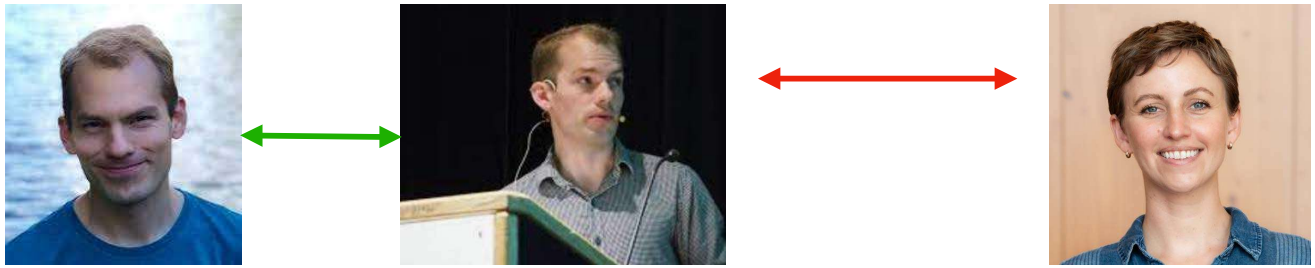


[He, Chen, Xie, et al. 2021]

Metric Learning

- Euclidean distance in input space may be not ideal
- Instead: learn a metric that respects desired properties

- Goal: learn a metric where:
  - data points that "belong together" are *similar* (close together)
  - data points that are "different" are dissimilar (far apart)
- "Supervision": similarity information.



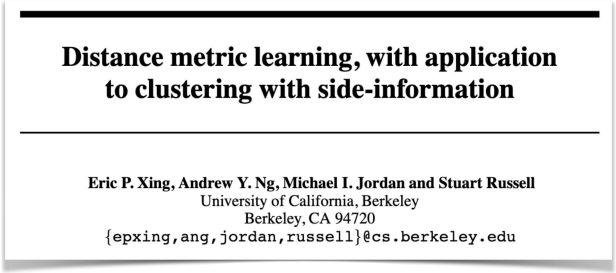
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"Losses": upper/lower bound constraints

- first approach *(Xing et al 2003)*:

$$\min_{A \succeq 0} \sum_{(i,j) \in S} d_A(x_i, x_j)^2 \qquad \text{min distance of similar points}$$

$$\text{s.t.} \sum_{(k,\ell) \in D} d_A(x_k, x_\ell)^2 \geq 1 \qquad \text{keep distance of dissimilar points}$$


Distance metric learning, with application to clustering with side-information

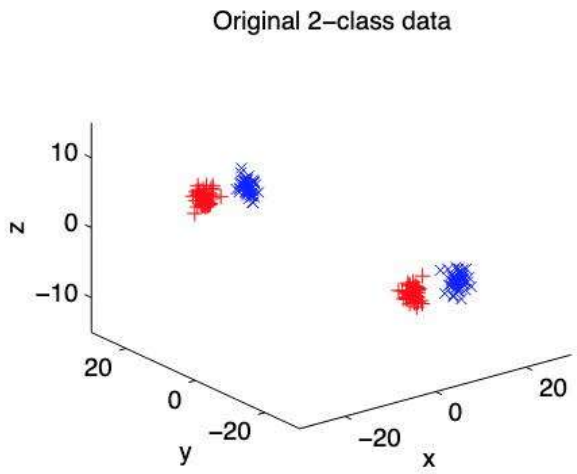
Eric P. Xing, Andrew Y. Ng, Michael I. Jordan and Stuart Russell
University of California, Berkeley
Berkeley, CA 94720
{epxing,ang,jordan,russell}@cs.berkeley.edu

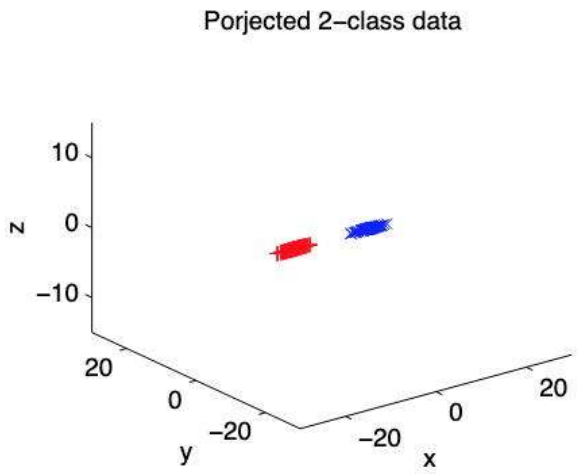
introduced the term and problem in 2003

- can swap objective and constraint (upper bound for similar pairs)

- many related ideas & follow-ups, e.g.
  information-theoretic metric learning (Davis et al 2007):
  preserve distribution information (relative entropy between Gaussians) while observing upper/lower bounds as constraints

Simple example





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figure: Ng et al 2003

Contrastive losses: intuition







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Contrastive losses

distance of dissimilar pair(s) **distance of similar pair(s)**

- Triplet loss *(Schroff et al 2015)*:

$$\mathcal{L}_{\text{triplet}}(\mathbf{x}, \mathbf{x}^+, \mathbf{x}^-) = \sum_{\mathbf{x} \in \mathcal{X}} \max\left(0, \|f(\mathbf{x}) - f(\mathbf{x}^+)\|_2^2 - \|f(\mathbf{x}) - f(\mathbf{x}^-)\|_2^2 + \epsilon\right)$$

margin

related: Large-margin Nearest Neighbor metric learning (LMNN) *(Weinberger et al 2009)*

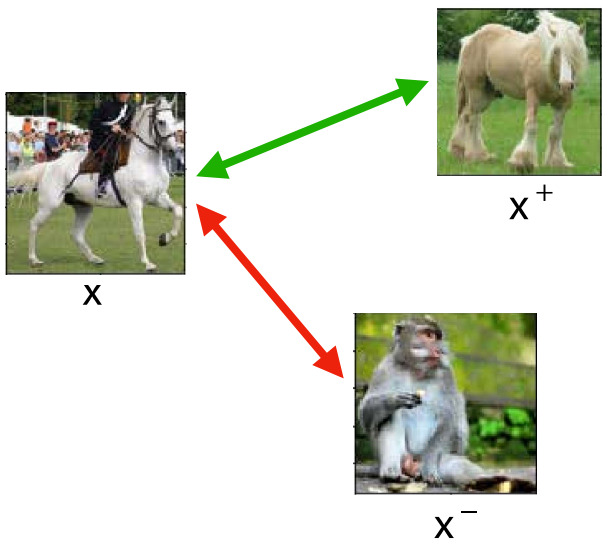
Contrastive losses

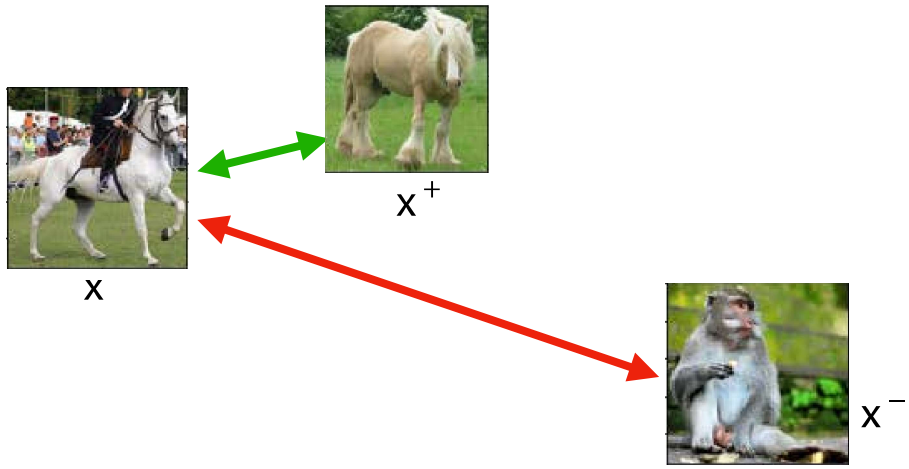
distance of dissimilar pair(s) **distance of similar pair(s)**

- Triplet loss *(Schroff et al 2015)*:

$$\mathcal{L}_{\text{triplet}}(\mathbf{x}, \mathbf{x}^+, \mathbf{x}^-) = \sum_{\mathbf{x} \in \mathcal{X}} \max\left(0, \|f(\mathbf{x}) - f(\mathbf{x}^+)\|_2^2 - \|f(\mathbf{x}) - f(\mathbf{x}^-)\|_2^2 + \epsilon\right)$$

margin





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Triplet network

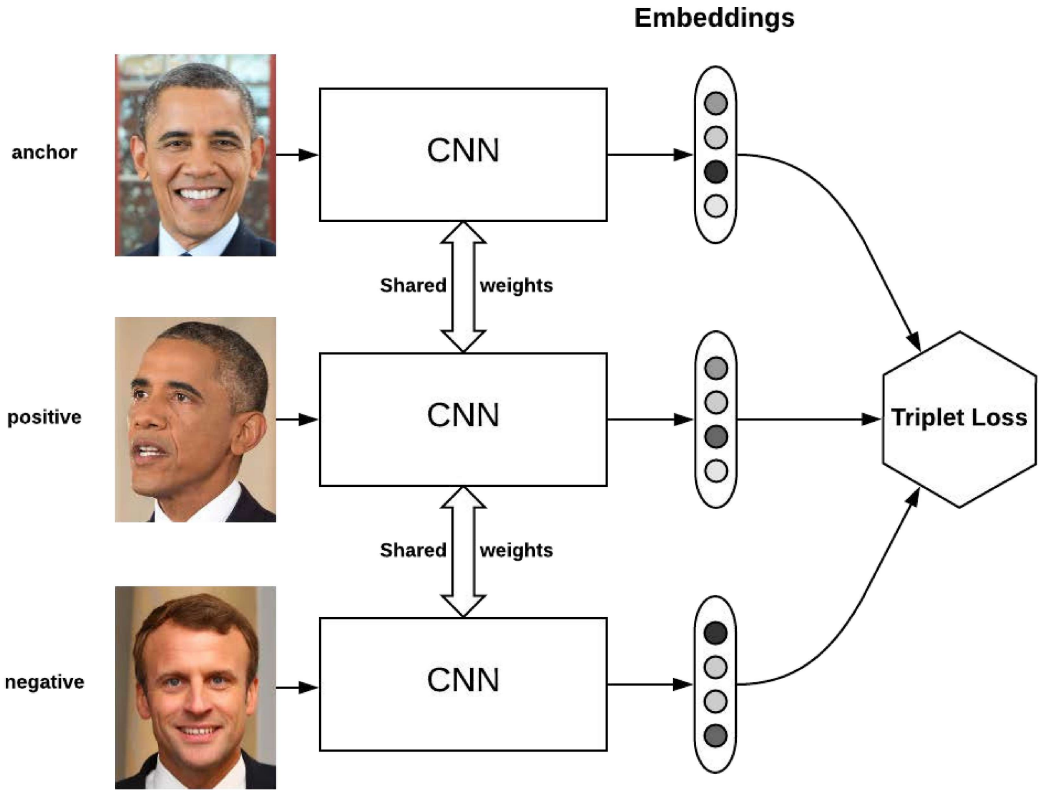


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figure: https://omoindrot.github.io/triplet-loss

Contrastive losses

distance of dissimilar pair(s) **distance of similar pair(s)**

- Improvements: compare to multiple negatives per positive pair, e.g. Lifted structured loss *(Song et al 2015)*: compare to all negatives in a batch

$$\mathcal{L}_{\text{struct}} = \frac{1}{2|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \max(0, \mathcal{L}_{\text{struct}}^{(ij)})^2$$

$$\text{where } \mathcal{L}_{\text{struct}}^{(ij)} = D_{ij} + \max\Big(\max_{(i,k)\in\mathcal{N}} \epsilon - D_{ik}, \max_{(j,l)\in\mathcal{N}} \epsilon - D_{jl} \Big)$$

$\|f(x_i) - f(x_j)\|_2$

or smooth relaxation of the max

Example embedding

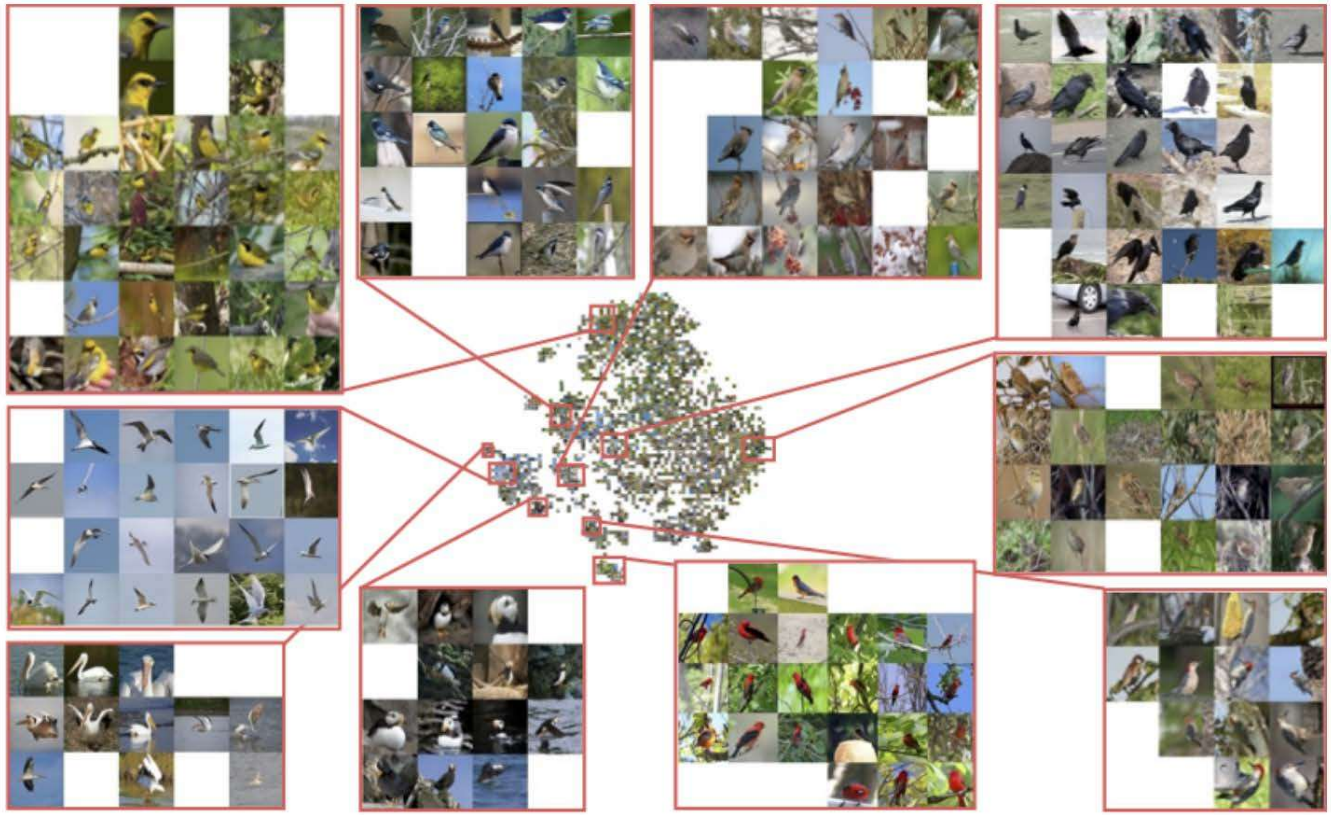
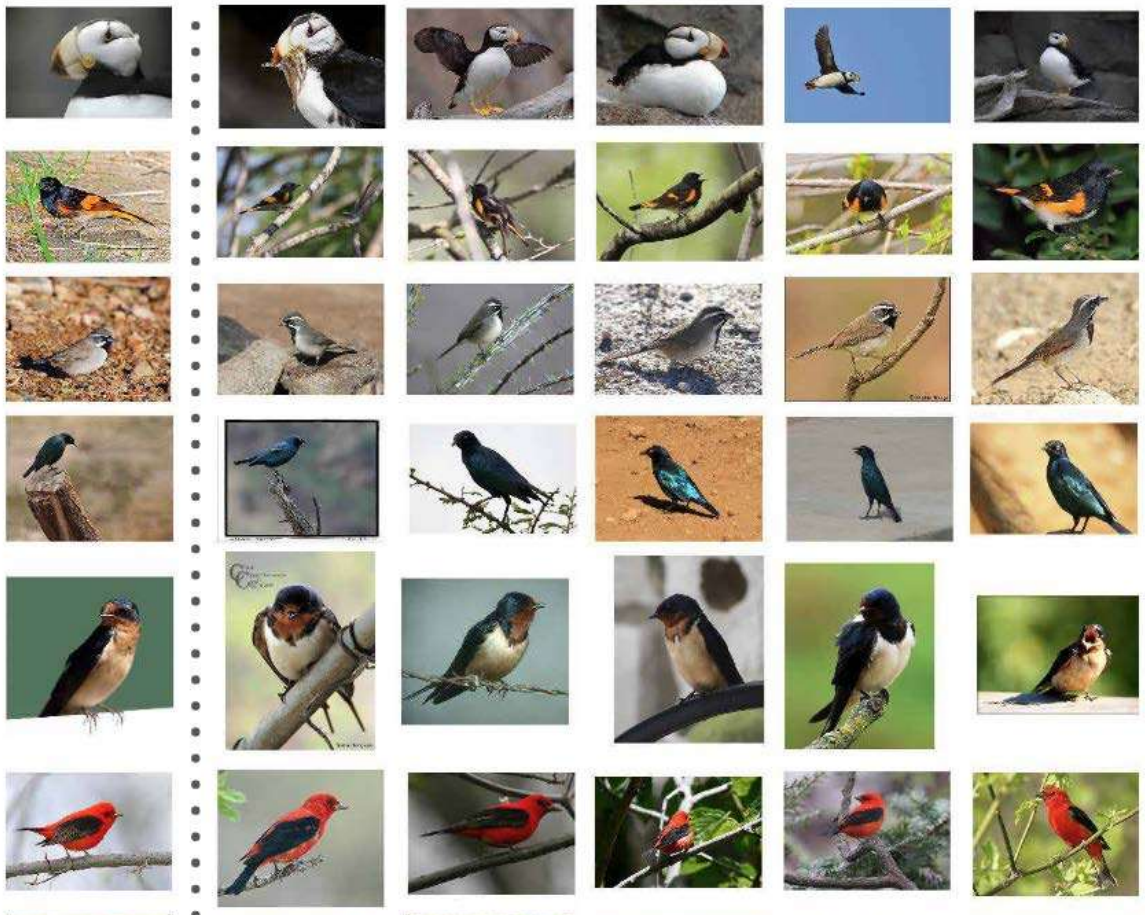


Figure 9: Barnes-Hut t-SNE visualization [36] of our embedding on the test split (class 101 to 200; 5,924 images) of CUB-200-2011. Best viewed on a monitor when zoomed in.

Courtesy of Song, et al. Used under CC BY-NC-SA.

figure: Song et al 2015

Example query results (neighbors)



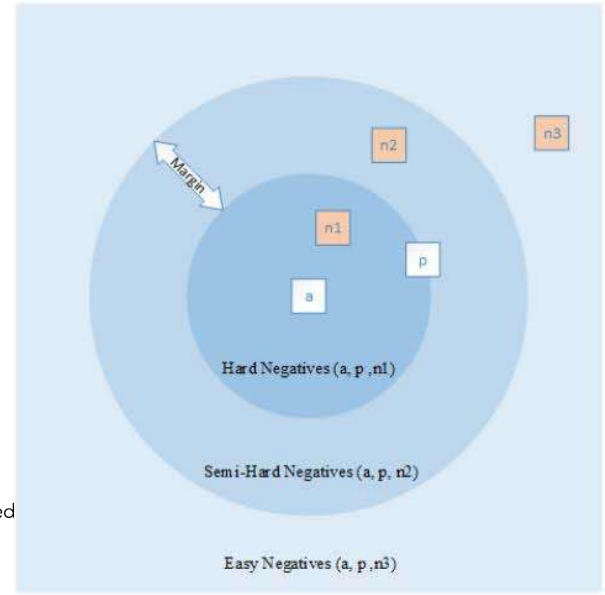
Courtesy of Song, et al. Used under CC BY-NC-SA.

figure: Song et al 2015

Which pairs should we present?

"hard" negatives:

- currently "misplaced", i.e., closer to anchor than a positive example
- accelerate learning, needed for triplet loss



Hard Negative Mining
$d(a,n) < d(a,p)$

Semi-Hard Negative Mining
$d(a,p) < d(a,n) < d(a,p) + margin$

Easy Negative Mining
$d(a,p) + margin < d(a,n)$







Fig 4. Courtesy of Kaya and Bilge. Used under CC BY
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Figure 4. Negative Mining.

Figure: Kaya & Bilge: Deep Metric Learning: A Survey, 2019

Roadmap: similarity-based representation learning

- Representation learning — why?
- What is a “good” representation?
- Metric learning
- Contrastive representation learning (self-supervised)
  - What does it do?
  - Models

Self-supervised contrastive representation learning

- Ideas from metric learning and self-supervision

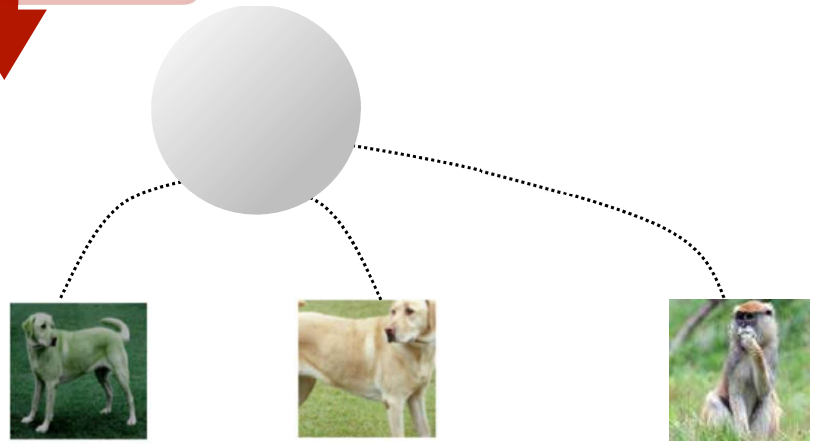
Common setup

- Encoder maps data onto a hypersphere: $f : \mathcal{X} \to \mathbb{S}^{d-1}$
- Cross-entropy for softmax "classifier"

$$\min_f \; \mathbb{E}_{(\mathbf{x},\mathbf{x}^+)\sim p_{pos},\{\mathbf{x}_i^-\}_{i=1}^N \sim p_{data}} \left[-\log \frac{e^{f(\mathbf{x})^\top f(\mathbf{x}^+)/\tau}}{e^{f(\mathbf{x})^\top f(\mathbf{x}^+)/\tau} + \sum_{i=1}^{N} e^{f(\mathbf{x})^\top f(\mathbf{x}_i^-)/\tau}} \right]$$

pull positive pair together

push negative pairs apart



- *Noise-contrastive estimation (NCE) (Gutmann & Hyvärinen 2010), InfoNCE loss (van den Oord et al 2018), … similar losses also in metric learning*

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Why map to a hypersphere?

- more stable training (logistic regression needs regularization)
- well-clustered classes on hypersphere are linearly separable (cut off caps)

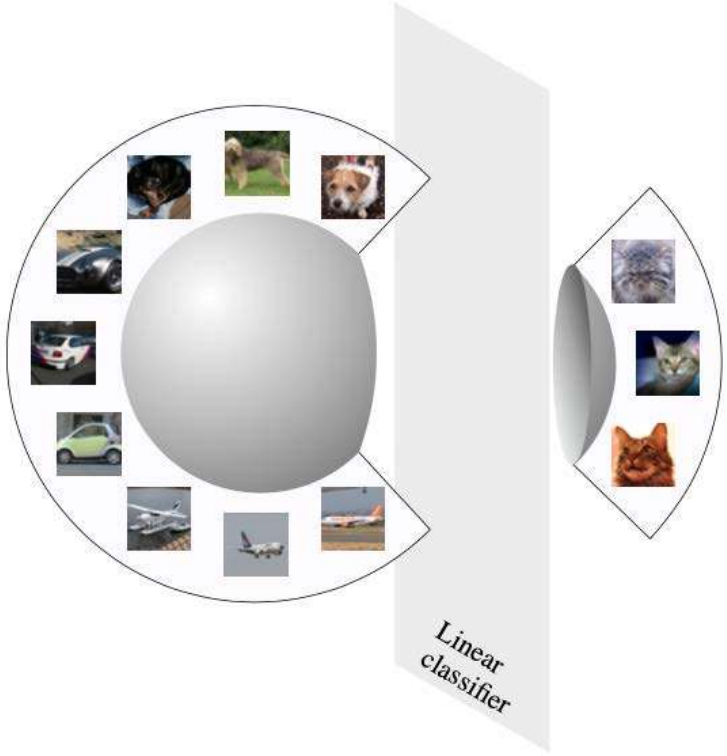


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figure: Wang & Isola 2020

How can we make this "self-supervised"?

$$\min_f \; \mathbb{E}_{(\mathbf{x},\mathbf{x}^+)\sim p_{pos},\{\mathbf{x}_i^-\}_{i=1}^N \sim p_{data}} \left[-\log \frac{e^{f(\mathbf{x})^\top f(\mathbf{x}^+)/\tau}}{e^{f(\mathbf{x})^\top f(\mathbf{x}^+)/\tau} + \sum_{i=1}^{N} e^{f(\mathbf{x})^\top f(\mathbf{x}_i^-)/\tau}} \right]$$

pull positive pair together

push negative pairs apart



- What are the similar (positive) and dissimilar (negative) pairs?

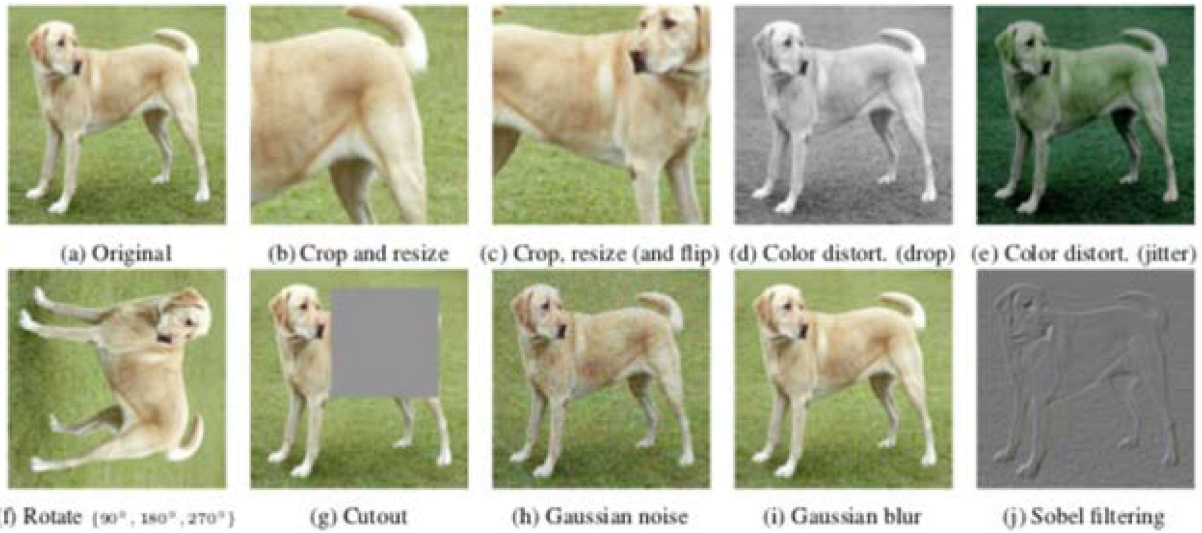
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What are positive and negative examples?

Negative examples:
randomly uniformly
drawn from data



Positive examples:
perturbations that keep
semantic meaning,
data augmentation



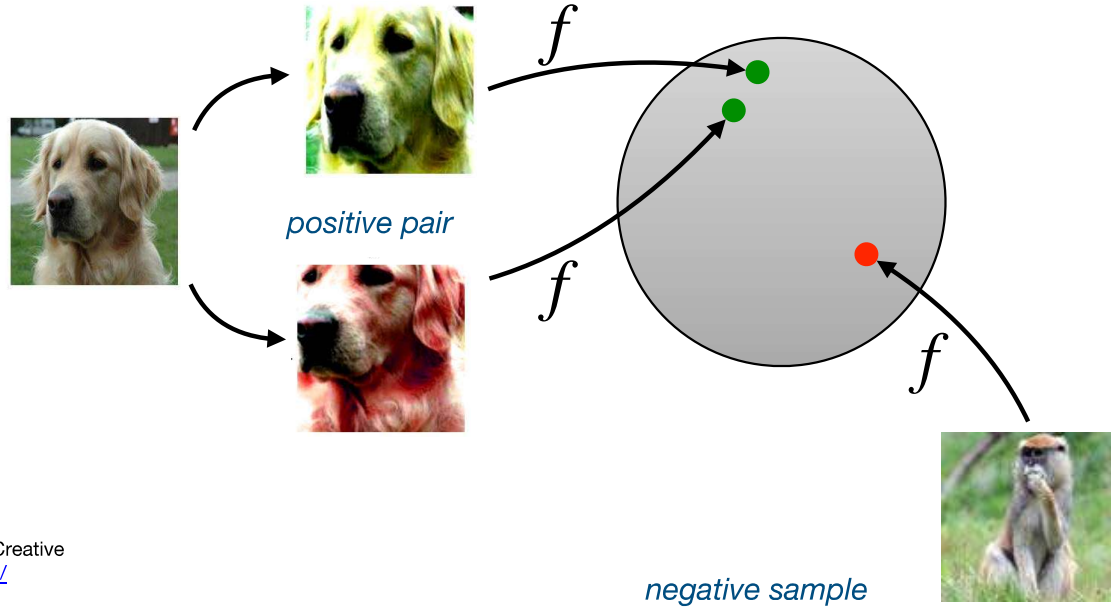
(Chen, Kornblith, Norouzi, Hinton 2020)

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Positive and negative samples

e.g. SimCLR:

- for each data point in the batch, generate 2 random augmentations as positive pair
- all other 2(B-1) augmented samples in the batch (of size B) are used as negatives



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What is the contrastive loss doing?

- Loss function encourages:

1. Concentration/Alignment: Data from the same class is close together, remove irrelevant information

2. Separation: classes are well separated, do not lose information

- Data encourages:

3. Robustness to **irrelevant** perturbations

Ingredients to make self-supervised CL work (better)

- heavy data augmentation
- projection heads
- large batch size (many negative examples)
- choice of data pairs / hard negative examples

} SimCLR model